Movie Recommendation System Using Machine Learning

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Abstract

With the rise of digital entertainment media, users are exposed to a wide range of movies, making it difficult for them to select content based on their preferences. The "Movie Recommendation System" helps users find suitable movies based on their similarity.

In this project, a content-based movie recommendation system is implemented using a CSV file containing movie information, including movie names, genres, keywords, and movie overviews. Natural Language Processing (NLP) and Machine Learning algorithms, such as TF-IDF Vectorization and Nearest Neighbors, are employed for movie recommendation.

Introduction

These are the platforms where recommendation systems are commonly implemented:

1.Netflix

2.Amazon Prime Video

3.YouTube

4.Spotify

Project Objective: The main objective of this project is to develop a recommendation system model that can perform the following tasks:

Read the movie information from a CSV file. Process the textual information related to the movies. Compute the similarity between the movies. Make a recommendation for the top similar movies

Objectives

- To preprocess movie dataset data.
- To apply NLP techniques on movie descriptions.
- To build a similarity-based recommendation model.
- To recommend movies based on user input.
- To evaluate recommendation quality.

Implementation Tools

Python

Pandas

NLTK

Scikit-learn

TF-IDF Vectorizer

Google Colab

Working of the System

1. User enters a movie name.
2. System searches movie index.
3. TF-IDF vectors compared.
4. Similarity scores calculated.
5. Top N movies recommended.

Results

The system successfully recommends movies based on textual similarity.

Observations:

- Movies with similar genres and storylines are recommended.
- Recommendation speed is fast after model training.
- Works without user history (content-based filtering).

Applications

- OTT platforms
- E-commerce recommendations
- Music streaming apps
- Online learning platforms

Conclusion

The Movie Recommendation System demonstrates how machine learning and NLP techniques can be applied to solve real-world problems. Using TF-IDF and similarity measures, the system effectively recommends movies based on content similarity, improving user experience and decision-making.

References

1. Scikit-learn Documentation — https://scikit-learn.org
2. NLTK Documentation — https://www.nltk.org
3. Kaggle Movie Dataset
4. Ricci, Rokach & Shapira — *Recommender Systems Handbook*

Google Colab LINK

https://colab.research.google.com/drive/1GWIngQLmRN6c9ZT2UqmZPWpsjNuYdDv4?usp=sharing